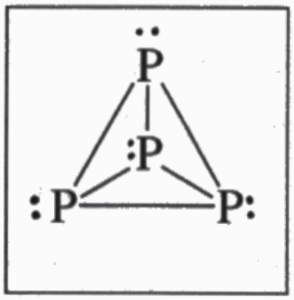
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 3

Part A



Part B

i) The entropy decreases because as the reaction progresses P_4 (s) will react with gas O_2 to form P_4O_{10} (s). Since there will be no gas as a product and the particles turned to solid, there is less distribution of molecules and less dispersion of molecules, decreasing entropy. & randomness

ii) The student's claim is correct. Based on the equation $\Delta G° = \Delta H° - T\Delta S°$, the reaction will be favorable if $\Delta H°$ is negative and $\Delta S°$ is positive. In this reaction, $\Delta H°$ is negative, but $\Delta S°$ is also negative, so the reaction favorability is only driven by enthalpy.

Question 3 is continued on the next page.

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Question 3

Part C

i) $q = mc\Delta T$

total mass = 100.0 g + 0.100 g = 100.1 g

$q = (100.1\ g)(4.18\ \frac{J}{g\,°C})(22.38°C - 22.00°C)$

$\rightarrow 159\ J \times \frac{1\ kJ}{1000\ J} =$ 0.159 kJ of heat is released

ii) $\Delta H°_{rxn} \rightarrow \frac{kJ}{mol\ rxn}$

Equation: $P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$

0.159 kJ/mol rxn 0.0265 kJ/mol rxn 0.0398 kJ/mol rxn

$\Delta H°_{rxn} = [0.0398\ kJ/mol\ rxn] - [0.0265 + 0.159] = -0.146\ kJ/mol\ rxn$

Part D

$q = mc\Delta T \rightarrow \Delta T = \frac{q}{mc}$

compared to the first trial, the ΔT for the second trial would be greater. Using the equation q=mcΔT, there is an inverse relation between mass and ΔT. So if mass decreases, ΔT increases

Question 3 is continued on the next page.

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Question 3

Part E

First equation: ~~divide to~~ multiply by $\frac{1}{4}$

second equation: remains the same

$\frac{1}{4} P_{4(s)} + \frac{6}{4} Cl_{2(g)} \rightarrow \cancel{PCl_{3(g)}}$ $\Delta H_1^\circ = (-1148 \text{ kJ/mol rxn}) \frac{1}{4}$

$+$ $\cancel{PCl_{3(g)}} + Cl_{2(g)} \rightleftarrows PCl_{5(g)}$ $\Delta H_2^\circ = -88 \text{ kJ mol rxn}$

$\frac{1}{4} P_{4(s)} + \frac{5}{2} Cl_{2(g)} \rightarrow PCl_{5(g)}$ $\boxed{\Delta H_f^\circ = -375 \text{ kJ/mol rxn}}$

Part F

i) ~~$K_p = \frac{(P_{PCl_3})^2}{}$~~ ~~$K_p = \frac{(P_{PCl_5})}{(PCl_3)(P)}$~~ $K_p = \frac{(P_{PCl_5})}{(P_{PCl_3})(P_{Cl_2})}$

$K_p = \frac{(1.00 \text{ atm})}{(1.00 \text{ atm})(1.00 \text{ atm})} = \boxed{1.00}$ ~~atm~~ $K_p = 1.00$

ii) If the temprature increases, then the value of K_p should ~~increase.~~ decrease because ΔH_c° is negative / exothermic.

Continue to Question 4.